Brhane Teamrat Gidey

Software Engineer | Backend & Full-Stack Development | AI-Enabled Systems

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Summary

Software Engineer with more than five years of experience building production software across healthcare, financial services, commerce, enterprise platforms, and international client projects. My background covers backend engineering, full-stack development, mobile applications, system integration, and AI-enabled software, allowing me to work across the complete development lifecycle from database and API design to user-facing functionality and deployment. I have delivered systems using Java/Spring Boot, Python/FastAPI, Laravel, JavaScript/TypeScript frameworks, and Flutter while working directly with clients and stakeholders to turn operational requirements into maintainable software. My recent work also includes LLM-based applications, retrieval systems, and automated code-evaluation pipelines, bringing modern AI capabilities into a practical software engineering foundation centered on reliability, testing, and clear system design.

Core Technologies: Java, Spring Boot, Python, FastAPI, Node.js, Laravel, React, Vue.js, PostgreSQL, MongoDB, Docker, AWS, REST APIs, Microservices, RAG, LLM Integration.

Professional Experience

Freelance Software Engineer — **December 2022 – Present**
Upwork & Freelancer.com — Remote / International

- Delivered web, mobile, backend, SaaS, and AI-integrated products for international clients, taking features from requirements clarification and architecture through implementation, testing, demonstration, deployment, and production support.
- Designed multi-tenant applications, REST APIs, database schemas, authentication and authorization flows, payment integrations, and cross-platform interfaces using Java/Spring Boot, Python/FastAPI, Laravel, React/Vue, Flutter, PostgreSQL, Docker, AWS, and GCP.
- Coordinated sprint planning, technical discussions, code reviews, release preparation, and client communication while maintaining **Upwork Top Rated status, a 100% Job Success Score, and five-star Freelancer.com feedback**.

Software Engineer — **March 2020 – September 2023**
Medco Technology Solutions — Addis Ababa, Ethiopia

- Developed healthcare and government software covering electronic medical records, laboratory operations, prescriptions, pharmacy workflows, reporting, billing, inventory, and administrative processes.
- Designed backend services, REST APIs, relational database structures, authentication workflows, frontend modules, validation rules, reports, and external-system integrations from business requirements.
- Diagnosed application and integration defects, supported regression testing and production releases, reviewed implementation decisions with teammates, and helped junior developers resolve technical blockers.

ICT Consultant — **September 2019 – March 2020**
Nayna Saving and Credit Cooperative — Ethiopia

- Assessed the organization's technology environment and advised management on infrastructure, software needs, information security, data backup, recovery planning, and operational continuity.
- Administered network and end-user systems, resolving connectivity, access, configuration, hardware, and software problems while supporting staff with day-to-day technical issues.
- Supervised two IT staff members, introduced clearer support procedures, and provided user training that improved internal issue handling and routine system operation.

Selected Projects

Integrated Electronic Medical Record & Laboratory Information System — **Java, Spring Boot, Vue.js, PostgreSQL, REST APIs**

- Developed a healthcare platform connecting patient registration, clinical records, laboratory requests, results, reporting, and role-based workflows so clinical and laboratory information could be managed within the same operational system.
- Implemented Spring Boot services, PostgreSQL data models, API integrations, validation rules, and user-facing functionality while participating in requirements analysis, testing, deployment, and production support.

Electronic Prescription & Pharmacy Management Systems — **Java, Spring Boot, Flutter, MySQL**

- Built functionality for creating, transmitting, validating, and tracking electronic prescriptions between healthcare providers and pharmacies with controlled access and traceable transaction history.
- Developed supporting pharmacy workflows for medicine inventory, order processing, interaction checking, online fulfilment, billing, and operational record management.

Food Ordering & Delivery Platform

Spring Boot, Flutter, Vue.js, PostgreSQL, Stripe

- Developed customer, restaurant/kitchen, courier, and administration applications around a shared order lifecycle covering menus, order placement, preparation, dispatch, delivery, and status tracking.
- Implemented backend order routing, role-based operations, payment processing, notifications, database persistence, and integration points required to coordinate the different applications.

Contract-Guided LLM Code Generation

Python, PyTorch, Hugging Face, PEFT/LoRA, Docker

- Built pipelines for generating, parsing, validating, and integrating formal preconditions and postconditions into code-generation workflows, including dataset preparation and model fine-tuning experiments.
- Automated execution-based evaluation across multiple coding models, running test cases and classifying logical errors, runtime failures, timeouts, interface violations, and successful solutions for systematic comparison.

Medical Knowledge & Healthcare AI Assistant

Python, FastAPI, BM25, FAISS, RAG, Docker

- Developed a medical question-answering application over trusted healthcare documents using hybrid keyword and dense retrieval, semantic search, evidence filtering, and re-ranking before response generation.
- Exposed the retrieval and generation workflow through a containerized FastAPI service with real-time querying, streaming responses, source-grounded answers, and checks for unsupported output.

Distributed Job Recommendation System

PySpark, FAISS, Neo4j, MongoDB, RAG

- Built a distributed processing pipeline using PySpark for candidate and job data, FAISS for semantic retrieval, and Neo4j for representing relationships among candidates, skills, and job requirements.
- Added retrieval-grounded explanations to recommendation results, allowing matches to reflect semantic compatibility and related skills rather than relying only on direct keyword overlap.

Education

M.Sc. in Artificial Intelligence and Computer Science — **Expected October 2026**
University of Calabria — Rende, Italy

- Studied software and intelligent systems through coursework in Agile Development of Software Systems, Intelligent Systems, Deep Learning, Big Data Analytics and Reasoning, Neuro-Symbolic AI, Machine Learning Optimization, and Statistics for Data Science.
- Conducting thesis work on *Contract-Guided LLM Code Generation*, combining formal software specifications, automated code generation, execution-based testing, model fine-tuning, and systematic error analysis.
- Completing thesis research at York University in Toronto, Canada, working with multiple code-generation models and programming benchmarks to investigate more reliable software generation.

B.Sc. in Computer Science and Engineering — **October 2014 – July 2019**
Mekelle University – Mekelle Institute of Technology — Mekelle, Ethiopia

- Built a strong foundation in data structures and algorithms, object-oriented programming, software engineering, database systems, computer networks, distributed systems, and embedded computing.
- Completed an undergraduate thesis on an *Emotion-Based Music Player*, developing real-time facial-expression recognition and connecting predicted emotions with music recommendation.
- Applied C/C++, Java, database, computer graphics, artificial intelligence, neural-network, and systems concepts through laboratory work and software-development projects.

Certifications

- **Spring MVC, Spring Boot and REST Controllers** — Coursera
- **Building Scalable Java Microservices with Spring Boot and Spring Cloud** — Coursera
- **DevOps on AWS: Code, Build, and Test** — Coursera
- **AWS Cloud Practitioner Essentials** — Coursera